

SYNTHETIC DEMONSTRATION — NOT FOR CLINICAL USE

ClearPlan synthetic review

Analytical phantom · 30 × 2 Gy

Scenario: publication-dual-layer · Generated: 2026-09-11 00:00 UTC

Active plan						
Plan	Created	Dose/Fx [Gy]	Total [Gy]	Fractions	Target	
Synthetic dual-layer VMAT	2026-09-11 00:00	2	60	30	PTV_60	

Delivery and modulation summary				
Total MU	MU / Gy per fraction	PAM [0-1]	Mean aperture [cm2]	Small-aperture fraction
900	450	0.318	77.537	0

PAM target: PTV_60 | MU beam weighting. Small aperture: < 4 cm2. Descriptive metrics.
 Plan normalization: 100 %. TPS setting.

Beam	MLC model	Layers	MU	Nominal MU/min	Geometry	Availability note
SYNTHETIC_AR01	Synthetic staggered dual-layer	2	450	800	available	Geometry supplied.
SYNTHETIC_AR02	Synthetic staggered dual-layer	2	450	800	available	Geometry supplied.

Nominal MU/min: plan setting, not measured delivery.

Plan quality metrics (PQM)

Template	Resolved structure	Objective	Rule	Goal	Variation	Achieved	Status
PTV_60	PTV_60	D98	>=	57 Gy	55 Gy	60.635 Gy	pass
PTV_60	PTV_60	D2	<=	64.2 Gy	66 Gy	62.821 Gy	pass
CTV_60	CTV_60	D98	>=	57 Gy	55 Gy	61.601 Gy	pass
SpinalCord	SpinalCord	D2	<=	20 Gy	25 Gy	9.723 Gy	pass
Parotid_L	Parotid_L	Dmean	<=	26 Gy	30 Gy	30.931 Gy	fail (error)
Parotid_L	Parotid_L	V30	<=	35 %	45 %	43.168 %	variation (warning)
Parotid_R	Parotid_R	Dmean	<=	26 Gy	30 Gy	9.567 Gy	pass

PTV quality | Conformity, gradient and homogeneity

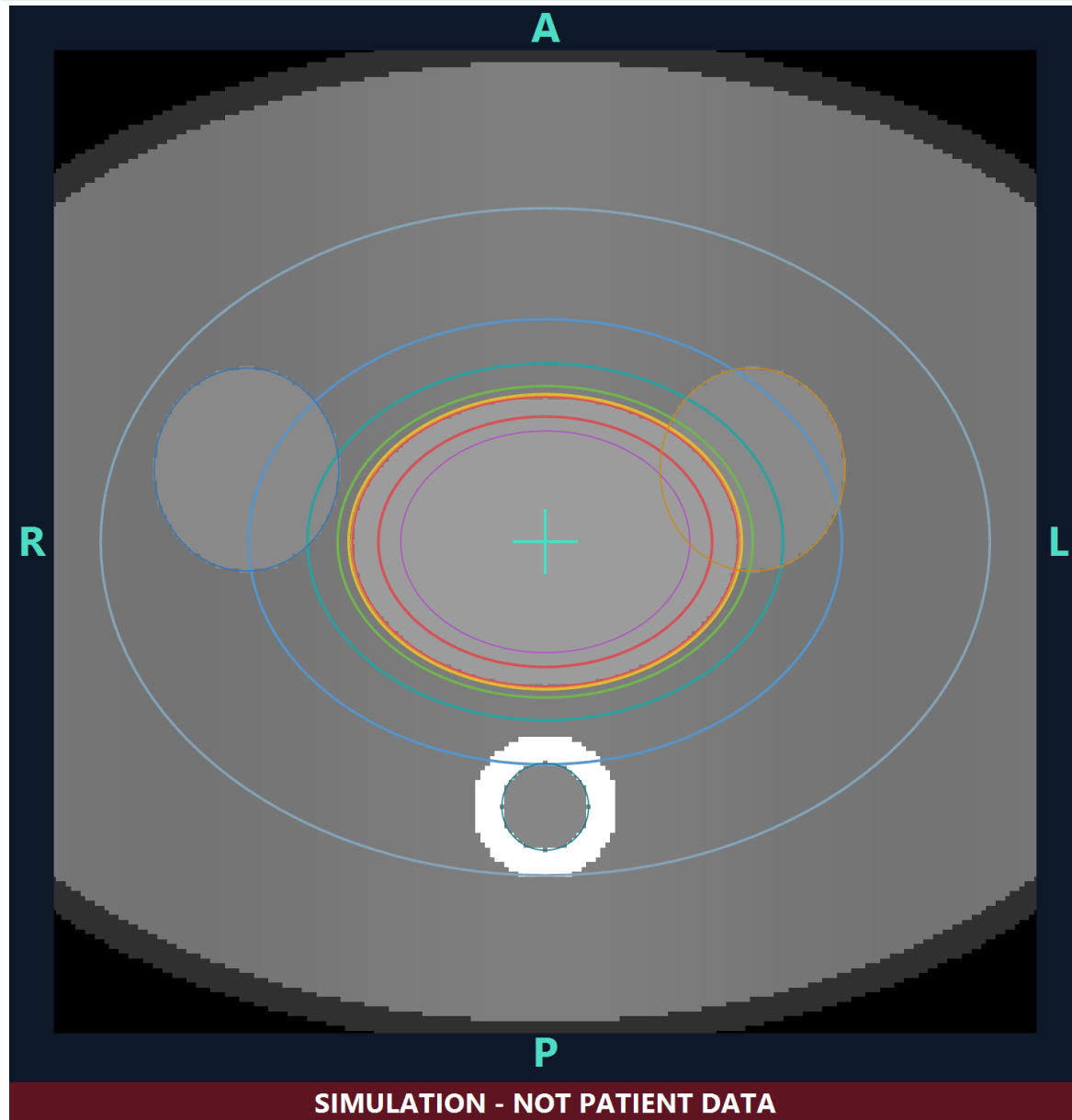
PTV	Plan Rx [Gy]	Paddick CI	1 / Paddick CI	GI V50/V100	HI (PlanCheck)
PTV_60	60	0.987	1.013	1.863	0.036

PTV_60: Synthetic analytical phantom; EXTERNAL whole-plan scope.

PTV / inputs	TV [cm3]	TV at Rx [cm3]	Body V100 [cm3]	Body V50 [cm3]	D2 [Gy]	D98 [Gy]
PTV_60	275.481	275.481	279.153	520.155	62.821	60.635

GI: whole EXTERNAL V50 / V100. HI: (D2 - D98) / plan Rx. Rx is the displayed plan prescription.

Synthetic CT | Transversal



Isocenter-centered · FOV 204 mm · W400 / L40 HU.

Contours: thin | Isodoses: thick

— PTV_60

— CTV_60

— SpinalCord

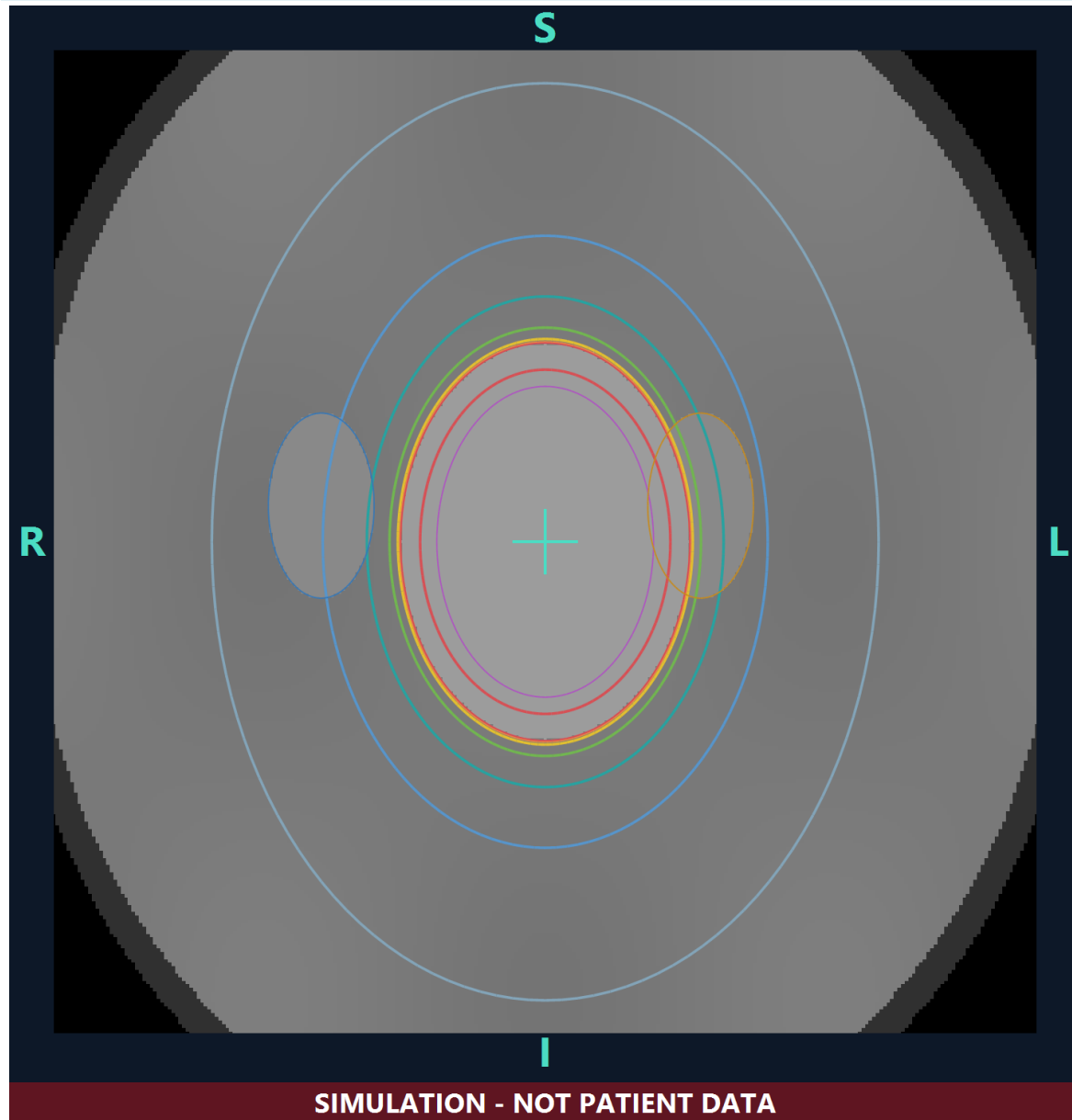
— Parotid_L

— Parotid_R

— 2% Rx / 1.2 Gy — 20% Rx / 12 Gy — 50% Rx / 30 Gy — 80% Rx / 48 Gy — 95% Rx / 57 Gy — 100% Rx / 60 Gy — 102% Rx / 61.2 Gy

Shared analytical phantom for CT, DVH and BEV; not dose calculated from apertures.

Synthetic CT | Coronal (orthogonal)



Isocenter-centered · FOV 272 mm · W400 / L40 HU.

Contours: thin | Isodoses: thick

— PTV_60

— CTV_60

— Parotid_L

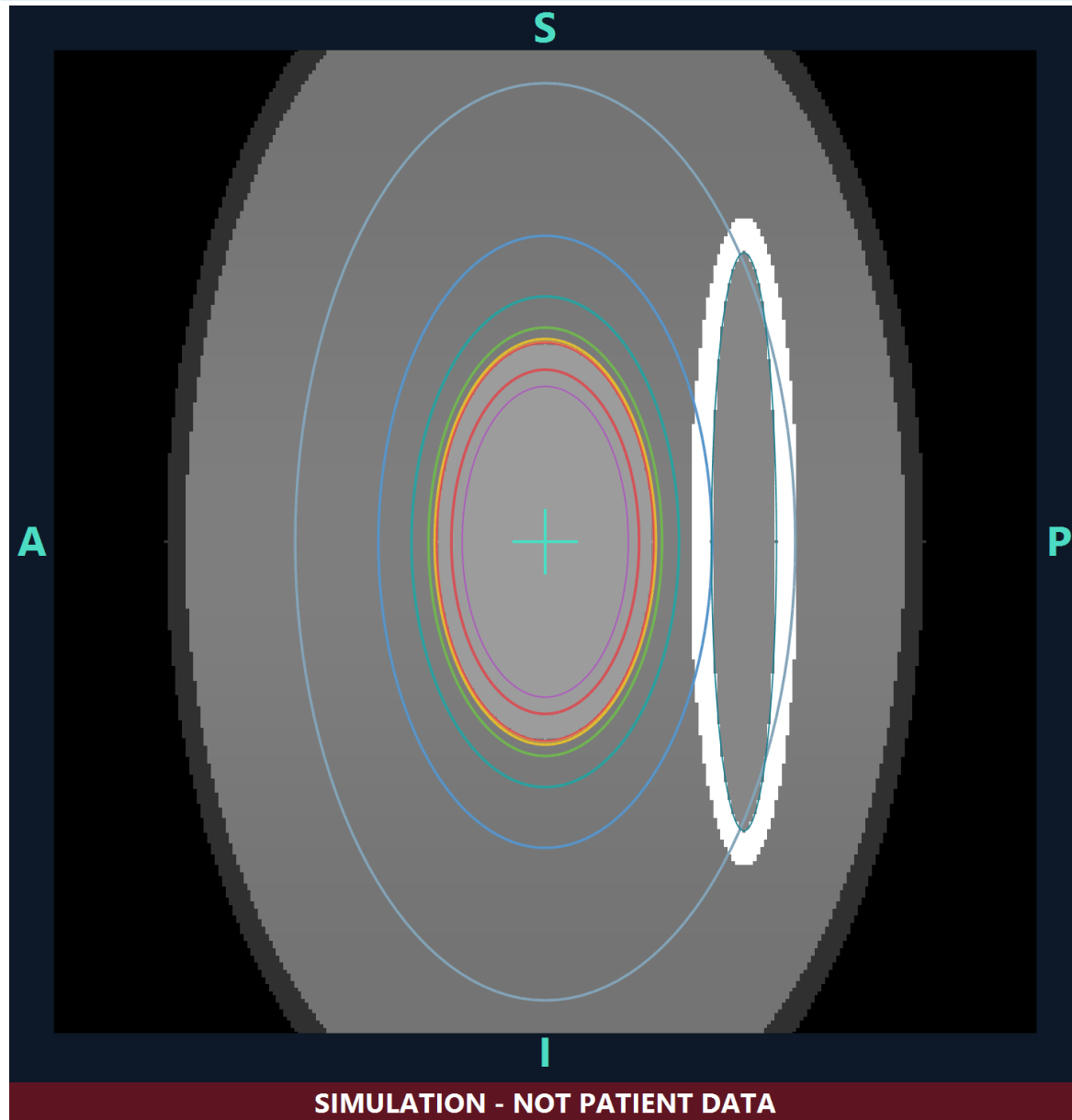
— Parotid_R

— 2% Rx / 1.2 Gy — 20% Rx / 12 Gy — 50% Rx / 30 Gy — 80% Rx / 48

Gy — 95% Rx / 57 Gy — 100% Rx / 60 Gy — 102% Rx / 61.2 Gy

Shared analytical phantom for CT, DVH and BEV; not dose calculated from apertures.

Synthetic CT | Sagittal



Isocenter-centered · FOV 272 mm · W400 / L40 HU.

Contours: thin | Isodoses: thick

— PTV_60

— CTV_60

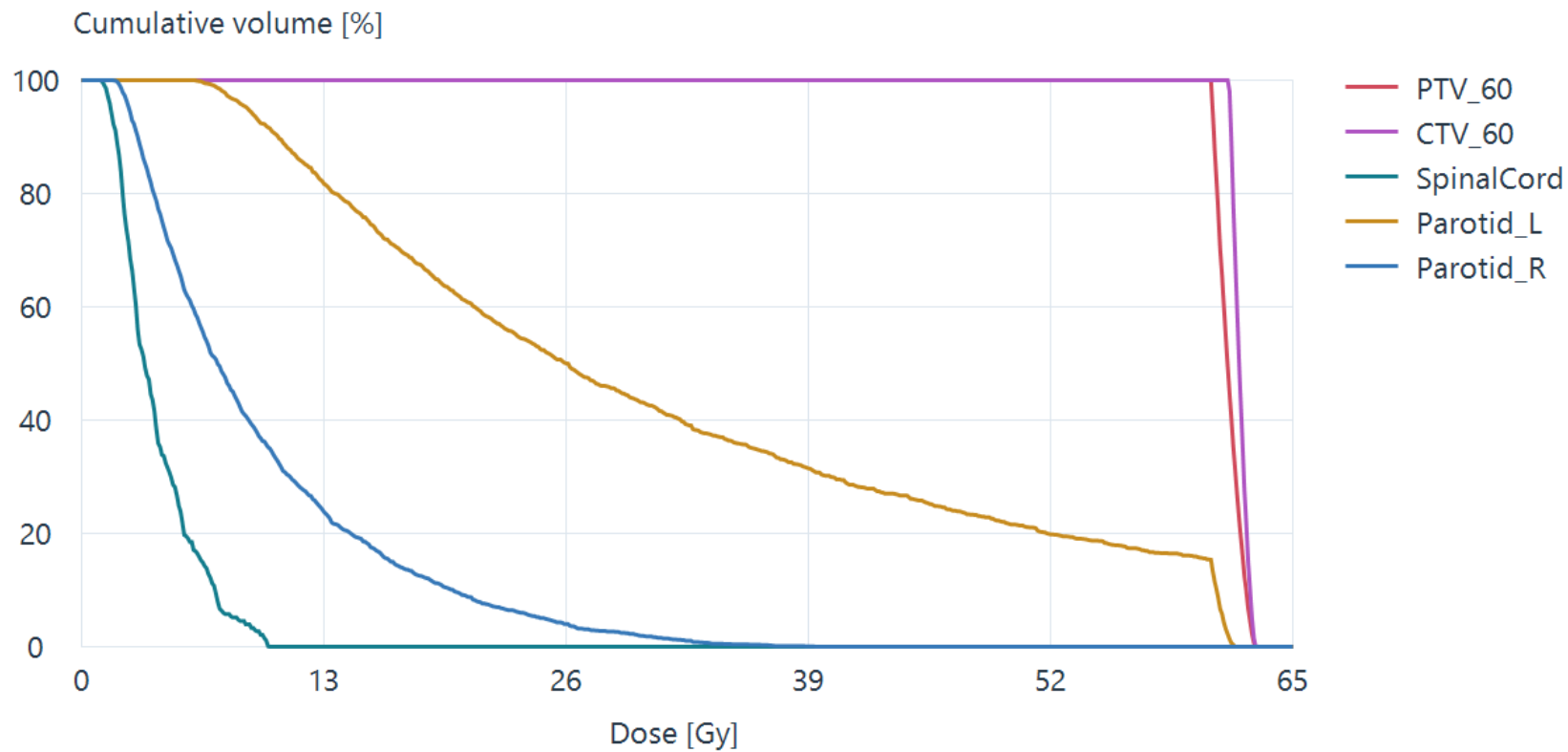
— SpinalCord

— 2% Rx / 1.2 Gy — 20% Rx / 12 Gy — 50% Rx / 30 Gy — 80% Rx / 48

Gy — 95% Rx / 57 Gy — 100% Rx / 60 Gy — 102% Rx / 61.2 Gy

Shared analytical phantom for CT, DVH and BEV; not dose calculated from apertures.

Dose-volume overview | Active plan



DVH statistics | Dose in Gy

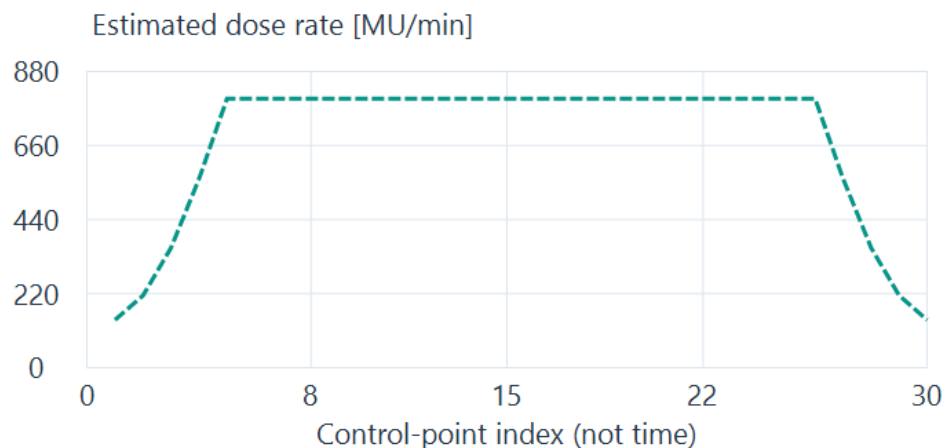
Structure	Volume [cm3]	Dmin / D98	Dmean / D50	Dmax / D2
PTV_60	275.481	approx. 60.6 / 60.635	approx. 61.563 / 61.49	approx. 63.1 / 62.821
CTV_60	124.389	approx. 61.5 / 61.601	approx. 62.155 / 62.111	approx. 63.1 / 62.892
Parotid_L	64.422	approx. 6.1 / 7.612	approx. 30.931 / 26.1	approx. 61.9 / 61.53
Parotid_R	64.422	approx. 1.7 / 2.184	approx. 9.567 / 7.29	approx. 39.6 / 29.928
SpinalCord	26.514	approx. 1 / 1.34	approx. 3.977 / 3.365	approx. 10.1 / 9.723

D98 / D50 / D2: dose at 98% / 50% / 2% volume. Approx.: synthetic estimate; -: unavailable. Selected curves are plotted.

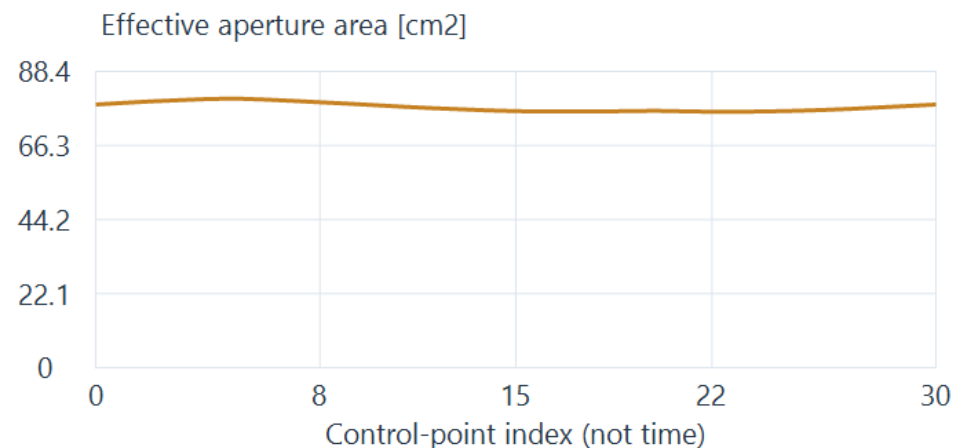
Control-point trajectories | Active plan

Nominal MU/min is a field setting, not a trajectory. Estimated plan trajectory: dashed, PlanCheck-style segment averages; supplied plan values: solid. Not measured delivery. Model excludes acceleration, leaf/jaw motion, ramping and holds; index is not time.

SYNTHETIC_AR01



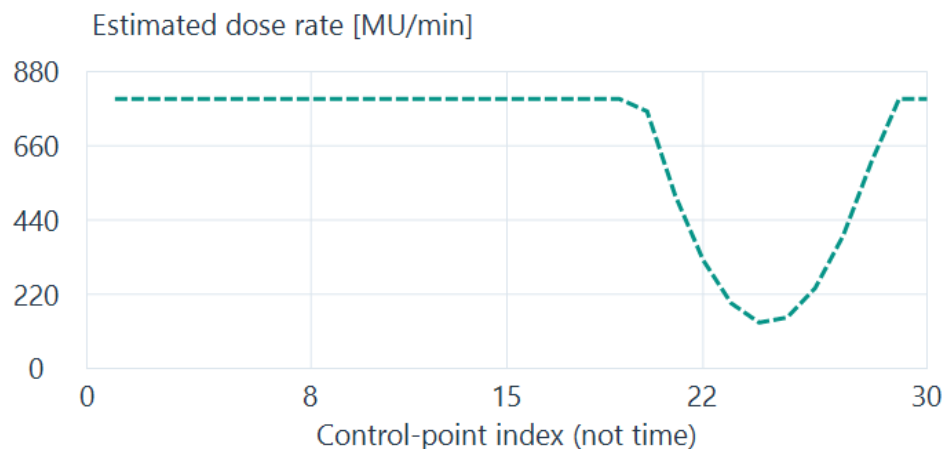
Dashed: estimated segment average, not measured



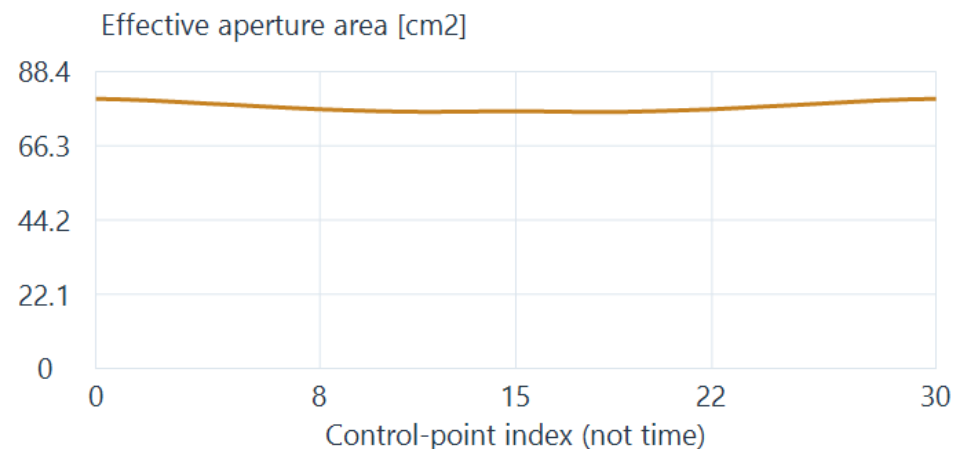
MLC layer intersection; jaws when present

Estimate profile: SYNTHETIC_DUAL_LAYER_RATE; gantry assumption: 12 deg/s; estimated duration: 36.2 s. Configured assumptions require local verification.

SYNTHETIC_AR02



Dashed: estimated segment average, not measured

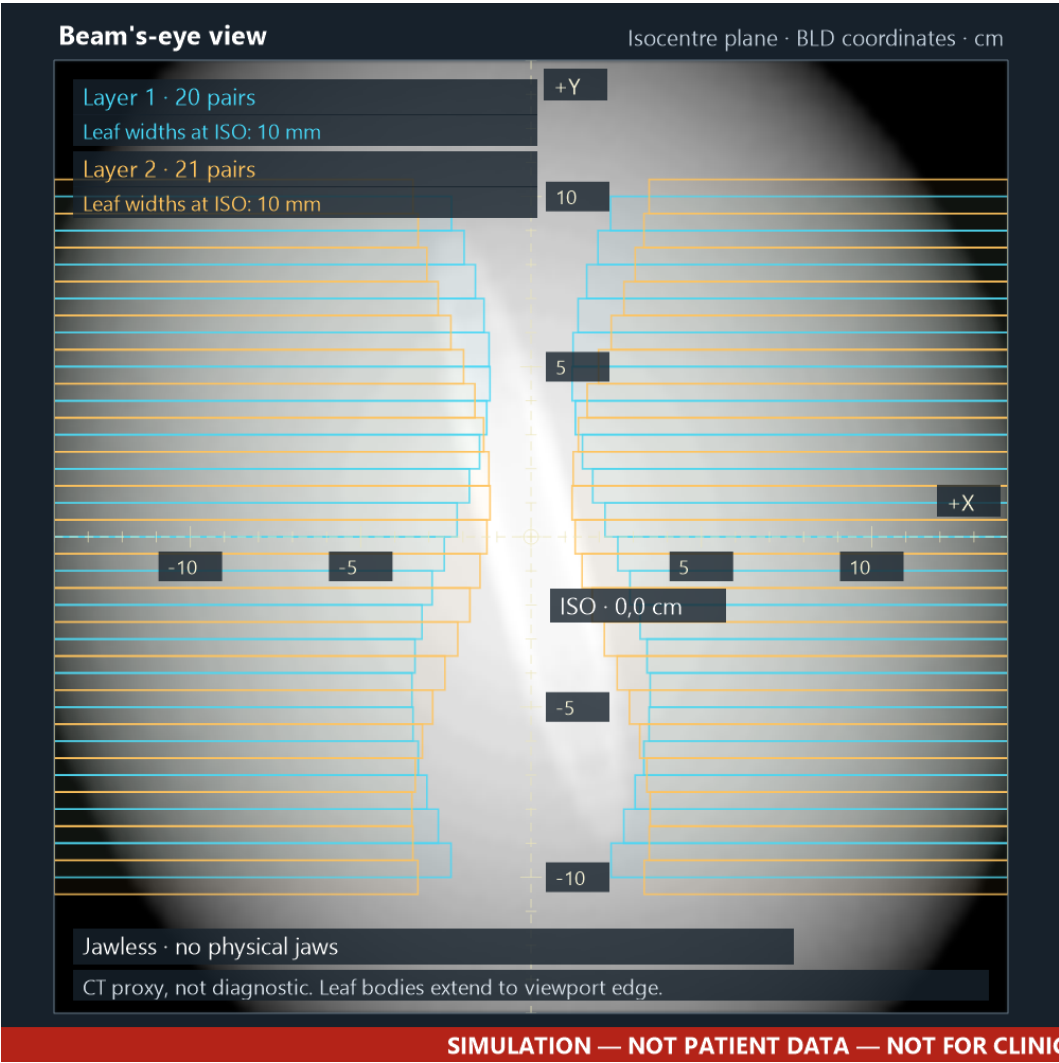


MLC layer intersection; jaws when present

Estimate profile: SYNTHETIC_DUAL_LAYER_RATE; gantry assumption: 12 deg/s; estimated duration: 36.2 s. Configured assumptions require local verification.

Beam's-eye view | SYNTHETIC_AR01

Field start · Exact CP 0 · Not arc-integrated fluence.



SYNTHETIC_AR01

Field start · CP 0 · Field 1

SYNTHETIC DUAL LAYER

Synthetic staggered dual-layer

Energy	6 MV FFF (synthetic)
Technique	VMAT
Beam meterset	450.0 MU
Aperture	Jawless · 2 MLC layers

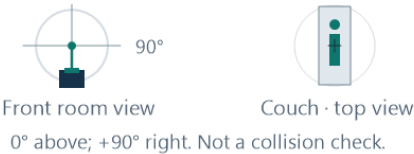
Angles	Current / field start
Gantry	180.0° / 180.0°
Collimator	15.0° / 15.0°
Couch	0.0° / 0.0°

Rate, planned	Unavailable
Rate, nominal	800 MU/min
Open area	78.62 cm ²

Synthetic CT projection available

Complete MLC / boundary geometry
 Control-point projection of the same analytical phantom as the three CT planes; synthetic HU...

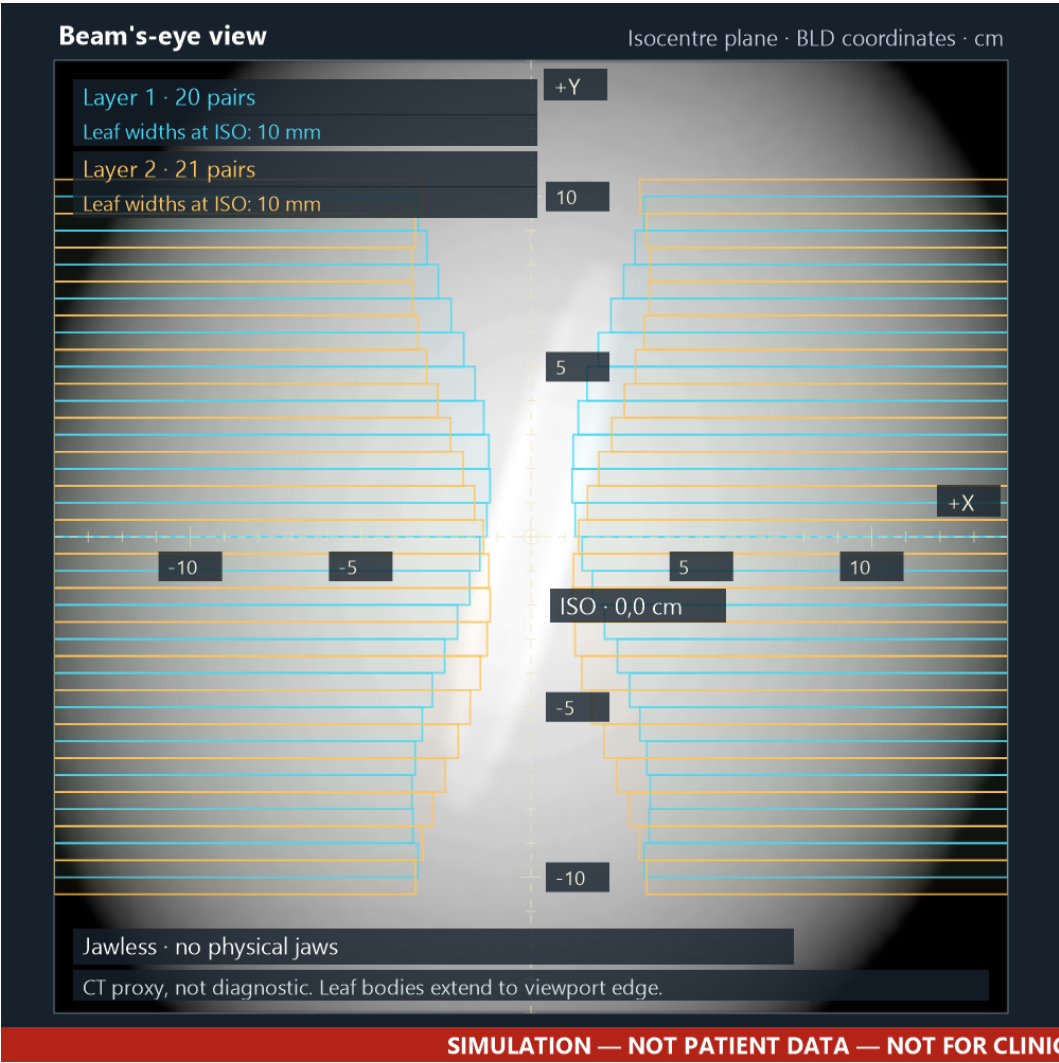
Orientation schematic



Field start · CP 0; gantry 180 deg; collimator 15 deg; couch 0 deg.

Beam's-eye view | SYNTHETIC_AR02

Field start · Exact CP 0 · Not arc-integrated fluence.



SYNTHETIC_AR02

Field start · CP 0 · Field 2

SYNTHETIC DUAL LAYER

Synthetic staggered dual-layer

Energy	6 MV FFF (synthetic)
Technique	VMAT
Beam meterset	450.0 MU
Aperture	Jawless · 2 MLC layers

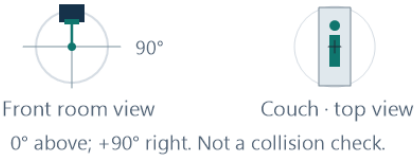
Angles	Current / field start
Gantry	0.0° / 0.0°
Collimator	345.0° / 345.0°
Couch	0.0° / 0.0°

Rate, planned	Unavailable
Rate, nominal	800 MU/min
Open area	80.37 cm ²

Synthetic CT projection available

Complete MLC / boundary geometry
 Control-point projection of the same analytical phantom as the three CT planes; synthetic HU...

Orientation schematic



Field start · CP 0; gantry 0 deg; collimator 345 deg; couch 0 deg.

Review details | Active plan

PlanCheck

Message	Status	Check	Category
Synthetic prescription: 30 fractions × 2 Gy = 60 Gy.	pass	synthetic-fractionation	Default
Injected synthetic finding: treatment approval is absent.	fail (error)	synthetic-approval	Default

Field identifiers and names

Order	Beam #	Current ID	Expected ID	Current name	Suggested name	ID status	Name status
1	1	SYNTHETIC_AR01	SYNTHETIC_AR01	180-0 UZ	180-0 UZ	pass	pass
2	2	SYNTHETIC_AR02	SYNTHETIC_AR02	0-180 UZ	0-180 UZ	pass	pass

Structure mapping

Template structure	Resolved structure	Status
PTV_60	PTV_60	pass
CTV_60	CTV_60	pass
SpinalCord	SpinalCord	pass
Parotid_L	Parotid_L	pass
Parotid_R	Parotid_R	pass